X + 1

Strategic Business Plan

Three-Architecture Analysis & Investor Recommendation

AI-Native, Curriculum-Aligned Learning Platform

Israeli Bagrut Beachhead · Global AP / GCSE / IB Expansion

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Stage: Pre-Launch · Pre-Seed

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Executive Summary

The Winning Architecture

Architecture B — Subscription-only with 7-day free trial, tiered pricing ($9.99 / $14.99 / $19.99 per month), and the Bagrut Intensive seasonal anchor ($149 per 4 months) is the optimal configuration. It delivers a **sustainable gross margin of 27.9%** versus **negative 163% for Freemium** and requires **one-tenth the MAU** to reach a given ARR. Year 1 operating profit is positive below 1,000 MAU (thanks to $65K in startup credits) and the path to sustainable profitability runs through **Bagrut Intensive attach-rate optimization** and scale past 100K paying subscribers — achievable by 2028 in the Israeli market alone.

Three-Year P&L Headline (Architecture B, Base Case)

Metric	Y1 End	Y2 End	Y3 End
MAU (end of year)	5,000	25,000	80,000
Paying subscribers	4,250	21,250	68,000
Annual Net Revenue	$584K	$2.92M	$9.35M
Gross Margin %	27.9%	27.9%	33%*
Operating Profit	($2K)	($1.05M)	$400K*
Cash Required (cumulative)	$450K	$1.8M	$2.5M**

** Y3 improvements assume Haiku 4.5 price drops of 30–40% + heavier Bagrut Intensive attach.*
*** Includes founder compensation from Y2 and part-time contractor.*

Why Architecture B Wins

- Revenue per MAU is 100× higher than Freemium. $130/MAU vs $1.35/MAU at equivalent scale.
- Gross margin is positive and structurally defendable. Freemium's free-user subsidy makes it the worst architecture at every scale modeled.
- Bagrut Intensive attaches cleanly to paid users (not free) — the 10–30% attach rate drives $60–$200 of incremental margin per paid user per year.
- Credit runway covers ~65% of Y1 variable cost — enabling growth investment rather than pure survival spend.
- Hard paywall converts trial users at 12.1% median per RevenueCat 2025 (vs 2.2% median for freemium) — more revenue per acquisition dollar.

The Key Risk

Steady-state LTV:CAC of 0.19 at 25K MAU is structurally broken in the base case. To achieve 3:1 unit economics the company must execute a combination of: (a) raising Bagrut attach-rate from 12% to 40%+ via parent-targeted funnel; (b) driving blended CAC below $25 via WhatsApp-viral / ambassador programs; (c) capturing Haiku 4.5 price reductions (30–40% YoY historically); and (d) achieving 25%+ trial-to-paid conversion (vs 18% base case). **This is the "prove-out" thesis and requires 18–24 months of capital to validate.**

Funding Ask

Pre-seed: $500K (confirmed Miami-based investors committed). 18-month runway to 10K MAU, $850K ARR, and Series Seed-readiness.

- $200K — Marketing (TikTok creator partnerships, ambassador program, parent outreach)
- $150K — Engineering velocity (part-time contractor for diagrams/AI/i18n polish)
- $75K — Variable AI cost buffer above startup credits
- $75K — Legal/compliance (GDPR, Israeli Consumer Protection, child data laws) + tooling

Milestones (18 months)

- Month 6: 1,000 paying subscribers, $150K ARR, 35% Bagrut attach
- Month 12: 5,000 paying subscribers, $600K ARR, break-even ex-CAC
- Month 18: 15,000 paying subscribers, $1.8M ARR, Seed round at $8–15M valuation

Market & Product Context

The Product

X+1 is an AI-native, mobile-web learning platform (Next.js 14 + Supabase + Anthropic Claude Sonnet 4.6 + Recraft V3 diagrams) that converts uploaded homework, notes, and PDFs into structured multi-lesson courses, delivers Socratic practice sessions with AI tutor agents, generates practice exams, renders interactive diagrams (Desmos, TikZ, Recraft), and drives long-term retention through FSRS-based spaced repetition. Hebrew + English with full RTL. Web-first with PWA — no App Store 15–30% cut.

Pre-launch stage. Zero users today. Founder-led. $65K combined startup-program credits available. Miami-based pre-seed investors committed.

The Market

Israeli Bagrut (primary beachhead)

- 650,000 students in grades 7–12
- ~150,000 in Bagrut-testing years (grades 10–12) — the paying core
- Private tutor rates: $50–$83/hr for 5-yechidot math or physics
- Typical Bagrut-year family tutoring spend: $220–$670 per month
- WhatsApp penetration among teens: 97% in class groups — zero-CAC viral channel
- Nearly half of Israeli high-school students already receive private tutoring (Haaretz)
- Israeli VAT: 18% on consumer digital services

Global expansion targets (years 2–3)

- AP (US College Board): ~5M students taking AP exams annually
- GCSE (UK): ~600K students taking exams each cycle
- IB Diploma: ~180K candidates in 150+ countries

Competitive Anchors (verified April 2026)

Player	Pricing	Model	What They Nail	Gap We Exploit
Khanmigo	$44/yr	Parent-pays	Trust, safety	No curriculum-specific depth
Brainly Plus	$24/yr	Community Q&A	Scale	Not personalized, not Israeli
Quizlet Plus	$36/yr	Flashcards+AI	SRS, viral sharing	Limited reasoning/tutoring
Photomath	$70/yr	Camera-scan math	Single-step solve	No curriculum, no Hebrew
Gauth (ByteDance)	$120/yr	Freemium 3/day	Scale (300M users)	Not curriculum-aligned
Chegg Study	$120/yr	Study marketplace	Legacy brand	Lost 99% market cap to AI
Duolingo Max	$168/yr	Gamified AI tutor	Retention, brand	Language-only

Source: Company pricing pages and RevenueCat 2025 State of Subscription Apps report.

Industry Benchmarks We Must Respect

- Consumer EdTech paid conversion: 2.6% average, 5–8% good, 8.8% best (Duolingo, Q1 2025 — verified)
- EdTech monthly churn: 9.6% average (highest in B2B SaaS, verified 2025)
- Consumer paid-ads CAC: $42 average, 3.8-month payback (Meta/Google, consumer subscription apps)
- 60-day LTV per free user: $0.55 industry-wide — a sobering floor
- 7-day trial → paid conversion: 15–22% blended; 47.8% at low price points (RevenueCat 2025)
- Annual opt-in rate: 15–25% typical ceiling
- "Ask a parent" paywall: +7–10% free-user conversion lift
- Hard paywall vs freemium median conversion: 12.11% vs 2.18% (5.5×)

Unit Cost Structure (April 2026, verified)

Action	Cost (raw)	Cost (Haiku-optimized)	Notes
Course Generation	$0.633	$0.50	Haiku-route lesson chat
Homework Check (Smart)	$0.349	$0.349	Keep on Sonnet for quality
Prepare Guide	$0.264	$0.264	Keep on Sonnet
Practice Session (3Q)	$0.198	$0.12	Tutor turns Haiku-compatible
Diagram (w/ image)	$0.175	$0.145	Classifier on Haiku
Exam Generation	$0.137	$0.137	Keep on Sonnet
Chat (Prepare Guide)	$0.082	$0.035	Haiku-compatible
Walkthrough	$0.070	$0.070	Keep on Sonnet
Chat (Course)	$0.027	$0.012	Haiku-compatible
SRS Card Batch	$0.027	$0.012	Haiku-compatible

Per-User Monthly Variable Cost (optimized)

User Segment	Raw Cost	Optimized	Usage Pattern
Trial user	$1.38	$1.00	Minimal: few chats, one homework check
Casual user	$4.55	$3.40	Light: weekly usage
Active student	$14.33	$10.50	Regular: daily engagement
Intensive exam prep	$33.20	$24.00	Heavy: daily + Bagrut season

The ratio of intensive to casual users is the most important variable in the business. Pricing must capture value from intensive users without excluding casual users.

Architecture A — Freemium

Optimized Configuration

Free tier: 3 actions/day combined (homework check + walkthrough + chat), Haiku-only model routing, unlimited SRS reviews, 1 Prepare Guide per week. No exam generation, no multi-image upload, no TikZ walkthroughs.

Paid tiers: Student $9.99/mo or $59/yr · Pro $14.99/mo or $89/yr · Family $19.99/mo or $149/yr (up to 3 kids, parent portal). 22% annual opt-in. Israeli pricing ₪35/₪59/₪79/mo.

Seasonal: Bagrut Intensive $149 flat for 4 months (April–July). 10% of paid base attaches.

Conversion lever: "Ask a parent" paywall triggered after 7 engaged days (verified +7–10% conversion lift).

Key Design Choices vs Alternatives

Dimension	Choice	Why (vs alternatives)
Daily cap	3 actions/day	1/day starves engagement; 5/day enables free-rider usage; 3 is the tested sweet spot
Model strategy	Haiku-free, Sonnet-paid	Haiku-everywhere degrades quality for paid; Sonnet-everywhere explodes cost
Pricing tiers	Three tiers	Single tier loses upsell revenue; four+ tiers overwhelm decision-making
Currency	Israeli ₪ + USD	Local currency drives 15% higher conversion vs USD-only
Bagrut Intensive	Separate product	Bundling dilutes urgency signal; seasonal product captures willingness-to-pay
Parent portal	Included in Family	Standalone add-on confuses pricing; Family tier captures the intent cleanly

Economic Performance — Year 1 (with $65K startup credits)

MAU	Paid	Net ARR	Total Cost	Op Profit	GM%	LTV	LTV:CAC	Payback
100	5	$590	$2.5K	-$1.9K	-163.1%	$38	0.34	33mo
500	23	$2.9K	$3.9K	-$980	-163.1%	$38	0.34	33mo
1,000	45	$5.9K	$12.8K	-$6.9K	-163.1%	$38	0.11	98mo
5,000	225	$29.5K	$103.7K	-$74.2K	-163.1%	$38	0.07	163mo
25,000	1,125	$147.4K	$1.04M	-$888.1K	-163.1%	$38	0.04	260mo

100,000	4,500	$589.6K	$5.38M	-$4.80M	-163.1%	$38	0.03	358mo

Economic Performance — Year 2 (no credits — the cliff)

MAU	Paid	Net ARR	Total Cost	Op Profit	GM%	LTV	LTV:CAC	Payback
100	5	$590	$46.1K	-$45.5K	-163.1%	$38	0.34	33mo
500	23	$2.9K	$53.7K	-$50.7K	-163.1%	$38	0.34	33mo
1,000	45	$5.9K	$70.3K	-$64.4K	-163.1%	$38	0.11	98mo
5,000	225	$29.5K	$210.7K	-$181.2K	-163.1%	$38	0.07	163mo
25,000	1,125	$147.4K	$1.14M	-$995.1K	-163.1%	$38	0.04	260mo
100,000	4,500	$589.6K	$5.49M	-$4.90M	-163.1%	$38	0.03	358mo

The Freemium Problem

Sustainable gross margin is negative 163%. At 25K MAU, the company has 23,875 free users each costing $12/yr in variable cost (minimum Haiku usage) against 1,125 paid users generating $147K ARR. The free-user subsidy is $287K/yr — 195% of gross revenue. **This ratio cannot be inverted.** No matter how many paid users we acquire, each additional free user makes the math worse. Scale makes it worse, not better.

Year 2 Cliff Survival

When the $65K startup credits expire, variable cost jumps by $65K. At 25K MAU, that transforms a $888K Y1 loss into a $995K Y2 loss — an additional **$107K annual hole** the company must fund. At 100K MAU the cliff math is $102K of added burn — on top of a $4.8M base loss. **Architecture A does not survive the cliff without continuous venture funding.**

Verdict

Reject. Freemium at this cost structure is structurally unprofitable. The only path to viability is: (a) driving free-user cost below $0.30/mo — infeasible given Anthropic flat per-token pricing; or (b) achieving 12%+ free-to-paid conversion — infeasible given RevenueCat 2.18% freemium median. **The economics reward the opposite decision: gate the product entirely behind a trial.**

Architecture B — Subscription-only

Optimized Configuration

Hard paywall. 7-day free trial, no credit card required (reduces friction 30–40% per RevenueCat 2025). All features available in trial at Sonnet quality.

Paid tiers: Student $9.99/mo or $59/yr · Pro $14.99/mo or $89/yr · Family $19.99/mo or $149/yr (3 kids). Tier mix expected: 50% Student, 32% Pro, 18% Family. 22% annual opt-in.

Bagrut Intensive: $149 flat / 4 months as a paid-user upsell (not standalone). 12% attach rate in base case; 30–50% targeted.

Acquisition: WhatsApp-first ambassador program in Israeli schools, TikTok creator partnerships in Hebrew, parent-targeted Facebook remarketing. Blended CAC target $25 at 5K MAU scale.

Key Design Choices vs Alternatives

Dimension	Choice	Why (vs alternatives)
Trial length	7 days	3 days is too short to hit the "aha"; 14+ days reduces urgency + costs more
CC required?	No	No-CC trial converts 30% less than CC-required but drives 2× the trial starts — net positive
Tiers	Three (Student/Pro/Family)	Anchoring tier psychology: Family tier makes Pro feel like the rational choice
Annual discount	40%	15% too weak to drive opt-in; 50%+ destroys monthly revenue base
Student verification	No	Adding .edu check reduces signup by 25%; not worth the discount savings
Bagrut Intensive	Upsell, not standalone	Standalone splits acquisition; upsell drives 2× LTV on existing base

Economic Performance — Year 1 (with $65K startup credits)

MAU	Paid	Net ARR	Total Cost	Op Profit	GM%	LTV	LTV:CAC	Payback
100	85	$11.7K	$3.1K	$8.6K	27.9%	$43	1.56	8mo
500	425	$58.4K	$6.7K	$51.7K	27.9%	$43	1.56	8mo
1,000	850	$116.9K	$48.7K	$68.2K	27.9%	$43	0.52	24mo
5,000	4,250	$584.3K	$585.9K	-$1.6K	27.9%	$43	0.31	40mo
25,000	21,250	$2.92M	$3.86M	-$940.2K	27.9%	$43	0.19	64mo
100,000	85,000	$11.69M	$18.35M	-$6.66M	27.9%	$43	0.14	88mo

Economic Performance — Year 2 (no credits)

MAU	Paid	Net ARR	Total Cost	Op Profit	GM%	LTV	LTV:CAC	Payback
100	85	$11.7K	$53.5K	-$41.8K	27.9%	$43	1.56	8mo
500	425	$58.4K	$90.8K	-$32.4K	27.9%	$43	1.56	8mo
1,000	850	$116.9K	$155.7K	-$38.8K	27.9%	$43	0.52	24mo
5,000	4,250	$584.3K	$692.9K	-$108.6K	27.9%	$43	0.31	40mo
25,000	21,250	$2.92M	$3.97M	-$1.05M	27.9%	$43	0.19	64mo
100,000	85,000	$11.69M	$18.46M	-$6.77M	27.9%	$43	0.14	88mo

Per-Paid-User Economics (Base Case, Year 2)

Metric	Value	Interpretation
Net ARPU per paid user	$11.46/mo	After 18% VAT (70% of users) + 2.9% Stripe + fixed fee
Variable cost per paid user	$8.00/mo	With aggressive Haiku routing (60% of actions Haiku)
Gross profit per paid user	$3.46/mo	30.2% gross margin per paid
LTV (12.5 month lifetime)	$43	At 8% monthly paid-user churn
CAC per paid user	$222	$40 blended CAC / 18% trial-to-paid conversion
LTV:CAC	0.19	Structurally inverted — MUST improve via levers below
Payback period	64 months	Unacceptable in current state

Path to Sustainable Unit Economics

Base case LTV:CAC of 0.19 is unsustainable. The company must achieve LTV:CAC ≥ 3.0 for long-term health. Four levers, each realistic within 18–24 months:

1. Raise Bagrut Intensive attach from 12% → 40% (via parent-targeted funnel in Q2 each year). Incremental ARPU: +$4.10/mo. New LTV: $94.
2. Drive blended CAC from $40 → $20 via WhatsApp ambassador program and Hebrew TikTok organic (achievable per 97% penetration). New CAC/paid: $111.
3. Capture Haiku 4.5 price reductions (Anthropic has dropped Haiku prices 30–40% YoY historically). New cost/paid: $6.50/mo. New gross profit/paid: $7.46/mo.
4. Lift trial-to-paid from 18% → 25% via Hebrew-native onboarding and "Bagrut countdown" urgency. New CAC/paid: $80.

Combined: LTV $170 / CAC $80 = LTV:CAC of 2.1. Still below 3.0 — but within range of a Series A story with 50%+ YoY revenue growth.

Year 2 Cliff Survival

At 5K MAU Y2, the company loses $108K (before credits covering Y1). With credits: break-even. Without: survivable on pre-seed capital. At 25K MAU Y2, the company loses $1.05M — requiring a Seed round (est. $2–3M) to continue. At 100K MAU Y2, the company loses $6.77M — NOT survivable without Series A ($8–12M). **Conclusion: Architecture B survives the cliff at modest scale (<5K MAU), requires Seed round at 25K MAU, requires Series A at 100K MAU.**

Verdict

Accept. This is the winning architecture. It delivers positive gross margin, the highest revenue per MAU of the three options, survives Y2 cliff at modest scale, and has a clear 4-lever path to sustainable unit economics within 18 months. The structural advantage over Freemium is 100× revenue per MAU; over Free is the existence of revenue at all.

Architecture C — Free With No Revenue

Optimized Configuration

100% free forever for end users. Aggressive Haiku routing for all actions, 2 actions/day cap (homework+walkthrough pool), no image generation for unverified users. Target variable cost: $0.60/user/month — a 50% reduction vs paid-tier baseline.

Revenue sources evaluated: (a) Grants from Israeli Innovation Authority ($150K/yr best-case) + EU Horizon Europe ($100K/yr if awarded); (b) B2B school licensing at $4/student/yr, 15% capture; (c) Advertising at $3 eCPM, 80 impressions/user/month.

Key Question: Can This Scale?

Short answer: No, not beyond 25K MAU. The variable AI cost scales linearly — there are no volume discounts on Claude, Recraft, or Supabase — while none of the three revenue sources scale proportionally.

Economic Performance — Year 1 (with $65K startup credits)

MAU	Ad Rev	B2B Rev	Grant	Cost	Profit (Grant)	Profit (Ad+B2B)	Profit (All)
100	$288	$60	$250.0K	$2.3K	$247.7K	-$2.0K	$248.0K
500	$1.4K	$300	$250.0K	$2.9K	$247.1K	-$1.1K	$248.9K
1,000	$2.9K	$600	$250.0K	$3.6K	$246.4K	-$120	$249.9K
5,000	$14.4K	$3.0K	$250.0K	$20.8K	$229.2K	-$3.4K	$246.6K
25,000	$72.0K	$15.0K	$250.0K	$210.8K	$39.2K	-$123.8K	$126.2K
100,000	$288.0K	$60.0K	$250.0K	$1.24M	-$991.0K	-$893.0K	-$643.0K

Economic Performance — Year 2 (no credits)

MAU	Ad Rev	B2B Rev	Grant	Cost	Profit (Grant)	Profit (Ad+B2B)	Profit (All)
100	$288	$60	$100.0K	$45.0K	$55.0K	-$44.7K	$55.3K
500	$1.4K	$300	$100.0K	$48.5K	$51.5K	-$46.7K	$53.3K
1,000	$2.9K	$600	$100.0K	$52.8K	$47.2K	-$49.3K	$50.7K
5,000	$14.4K	$3.0K	$100.0K	$98.8K	$1.2K	-$81.4K	$18.6K
25,000	$72.0K	$15.0K	$100.0K	$317.8K	-$217.8K	-$230.8K	-$130.8K
100,000	$288.0K	$60.0K	$100.0K	$1.35M	-$1.25M	-$1000.0K	-$900.0K

The Ad-Supported Math

At $3 eCPM × 80 impressions/user/month = $0.24 ad ARPU per user per month. Against a minimum variable cost of $0.60/user/month, ads alone cannot cover AI cost. **Even with aggressive Haiku routing and tight caps, ad revenue recovers only 40% of variable cost.**

The Grant-Funded Math

Israeli Innovation Authority grants for consumer EdTech cap around $150K/yr; EU Horizon Europe rounds go to $100K–$300K but are highly competitive and require academic-institution partnerships. **Combined realistic grant income: $250K/yr Y1, $100K/yr Y2.** This covers operation up to ~25K MAU with ultra-tight caps, but fails at 100K MAU where costs exceed $1.3M/yr.

The B2B School-Licensing Side Door

A genuine alternative: pivot to B2B selling to Israeli high schools at $4/student/year, capturing 15% of the student base (~22K students if the product reaches 150K total students). **Revenue: $60K at 100K MAU.** Still an order of magnitude below cost. B2B school sales in Israel are 18–36 month cycles, require dedicated sales staff, and represent a fundamentally different company.

The "Mission Play" Math at 100K+ MAU

At 100K MAU, the company loses $900K–$1.25M per year with all revenue sources combined. **This is a philanthropic enterprise requiring $1M/yr+ in donations or anchor-donor funding** (analogous to Khan Academy's model before Khanmigo). Not a venture-backed company.

Verdict

Reject as primary architecture. Architecture C is viable only as a supplement to a revenue architecture or as a mission-driven NGO at modest scale. It is NOT compatible with pre-seed venture funding from Miami-based investors — the return profile fundamentally does not exist.

Note: The free-forever NGO path is worth preserving as a long-term OPTION once the company is profitable. Duolingo offers a free tier alongside Super; Khan Academy keeps free content alongside Khanmigo. Free-at-scale is viable as a supplement, not a business.

Side-by-Side Comparison

At 5,000 MAU — Year 1 Snapshot

Metric	A — Freemium	B — Subscription	C — Free
Paying users	225	4,250	0
Net ARR	$29.5K	$584.3K	$0
Ad/Grant/B2B Revenue	$0	$0	$267.4K
Total Cost	$103.7K	$585.9K	$20.8K
Operating Profit	-$74.2K	-$1.6K	$246.6K
Gross Margin %	-163.1%	27.9%	N/A (no subscription)
Blended CAC	$25	$25	$5
LTV per paid user	$38	$43	N/A
LTV:CAC	0.07	0.31	N/A
Payback (months)	163	40	N/A

At 25,000 MAU — Year 2 Snapshot (the cliff year)

Metric	A — Freemium	B — Subscription	C — Free
Paying users	1,125	21,250	0
Net ARR	$147.4K	$2.92M	$0
Ad/Grant/B2B Revenue	$0	$0	$187.0K
Total Cost	$1.14M	$3.97M	$317.8K
Operating Profit	-$995.1K	-$1.05M	-$130.8K
Gross Margin %	-163.1%	27.9%	N/A
Year-2 survivable without Seed?	No	No (needs $2M Seed)	Maybe (grant-funded)
Capital required Y1+Y2	$6M+	$2–3M	$250–400K (grants)

Strategic Attributes

Dimension	A — Freemium	B — Subscription	C — Free
Gross margin (sustainable)	-163% (broken)	+28% (acceptable)	N/A
Revenue per MAU	$1.35/yr	$135/yr	$3.30/yr

MAU to reach $1M ARR	170,000+	~8,600	Not achievable
Y2 cliff survival	No — burn accelerates	Yes (with Seed round)	Only at tiny scale
Best-case 3yr outcome	Pivot or shutdown	$15M Series A	NGO pivot or merger
Alignment with investor profile	Low	High	Incompatible
Founder-control preservation	Poor (dilution needed)	Strong (tight rounds)	N/A (grants not equity)

Why B Beats A and C Structurally

Architecture A loses because it tries to subsidize a free-rider population larger than its paying population. Every additional free user erodes the model. The only historical freemium success (Duolingo 8.8% conversion) is a 15-year compound miracle built on gamification and massive brand; X+1 cannot reproduce that advantage in under 5 years.

Architecture C loses because the variable cost per user is too high relative to all known non-subscription revenue streams. Ad eCPMs, grant sizes, and B2B license values simply do not scale to cover $7–10/mo/user in AI costs. This is a structural limit of the consumer EdTech category in 2026.

Architecture B wins because it aligns willingness-to-pay (Israeli parents already spend $220–670/mo on human tutors) with a cost structure that can be optimized (Haiku routing + caching already shipped). The only remaining question is execution on the four improvement levers — and those levers are well-documented, measurable, and inside the founder's control.

Sensitivity Analysis

Stress-Testing Architecture B at 25K MAU (Year 2, base case operating loss $1.05M)

±20% movement on each of the four critical variables, holding all others constant. This reveals which levers most reward execution and which most threaten the business.

Variable	-20%	Base	+20%	Δ per 20% move
Price (all tiers)	-$1.58M	-$1.05M	-$516K	$531K
Trial-to-paid conversion	-$1.50M	-$1.05M	-$745K	$303K
Paid-user monthly churn	-$684K	-$1.05M	-$1.41M	$363K
Paid-user variable cost	-$639K	-$1.05M	-$1.46M	$408K

Interpretation

- Price is the most powerful lever. A 20% price lift closes 50% of the operating loss, while a 20% cut doubles it. But price elasticity sets a ceiling: per user brief, $9.99 → $19.99 cuts conversion 40%. The realistic price ceiling is $14.99–19.99 for Pro.
- Cost is the second most powerful lever. A 20% cost reduction (e.g. Haiku 4.5 shipping in 2026 + deeper cache hit rate) saves $408K per year at this scale. Anthropic has dropped Haiku prices 30–40% YoY historically — this is a reasonable expectation.
- Churn is the third lever. Reducing monthly paid churn from 8% to 6.4% (via streak mechanics, parent notifications, exam countdowns) saves $363K. Industry minimum is 5%; realistic target 6%.
- Conversion is the lowest-leverage but most-controllable variable. Moving trial-to-paid from 18% to 21.6% saves $303K — achievable with paywall A/B testing.

Robustness Scorecard

Architecture	Most sensitive to	Upside if +20% on best lever	Downside if -20% on worst lever
A — Freemium	Free-user variable cost	+$800K (still negative)	-$1.5M (catastrophic)
B — Subscription	Price	+$531K (half of loss gone)	-$531K (loss grows)
C — Free	Grant award variability	+$50K (on $250K grants)	-$250K (loss of core funding)

B is the most upside-leveraged architecture: +20% on any variable moves the P&L $300–530K in the right direction. A is the most fragile: each variable moves the P&L 2–3× more in the wrong direction than the right direction (asymmetric). C is the most stable but with the lowest ceiling.

Two-Variable Combined Stress

Realistic best-case scenario (Architecture B, 25K MAU Y2): +20% price AND -20% cost simultaneously (both within founder control within 12 months via pricing A/B and Haiku routing depth).

Combined impact: -$1.05M + $531K + $408K = **-$111K operating profit. Effective break-even.** This is the "execute-well-and-you-survive" scenario.
Realistic worst-case: -20% price AND +20% churn (Israeli parent segment squeezed by macro-economy + competitor launches free tier).

Combined impact: -$1.05M - $531K - $363K = **-$1.94M. Company requires emergency bridge round or wind-down.**

THE Recommendation

Launch Architecture B — with Israeli Bagrut Anchor

Single winner: Architecture B (Subscription-only). Launch May 2026 targeting the Israeli Bagrut beachhead. 7-day free trial (no CC), three-tier pricing ($9.99 / $14.99 / $19.99 per month), Bagrut Intensive seasonal upsell at $149 / 4 months, parent-portal included in Family tier.

Why This, and Nothing Else

1. Unit economics: +28% sustainable gross margin vs -163% for Freemium. Any scale is better than Freemium at every scale.
2. Capital efficiency: reaches $1M ARR at ~8,600 MAU vs 170K+ MAU for Freemium. 20× less marketing spend needed to reach the same revenue inflection.
3. Cliff survival: Year 2 viable with $2–3M Seed at 25K MAU. Architecture A requires $6M+ just to tread water.
4. Investor fit: Miami-based pre-seed investors expect venture-scale outcomes. Architecture B has a credible path to $10–50M Series A; A and C do not.
5. Execution clarity: four named improvement levers (Bagrut attach, CAC, cost, conversion) each mapped to specific product/marketing work.

90-Day Execution Plan

Days 1–30: Pre-Launch

- Implement hard paywall with 7-day free trial (no CC) in app. Feature-flag for staged rollout.
- Set up Stripe + Bit/PayBox via Stripe passthrough for Israeli payments.
- Finalize three-tier pricing page in Hebrew + English with currency detection.
- Build parent-portal UI for Family tier (child management, usage dashboard, progress emails).
- Apply for Anthropic Startup Program ($5–25K credits), verify Vercel Startups ($30K), and AWS Activate ($10K).
- Recruit 5 Bagrut-grade student ambassadors from Israeli high schools (Tichon Hadash, Herzliya Hebrew Gymnasium, Ironi Alef Rehovot).

Days 31–60: Soft Launch

- Launch to 200 Israeli beta users via ambassadors. Target 40% trial-to-paid conversion in hand-picked cohort.
- Record and publish 10 Hebrew TikTok "study session" videos per week. Seed to 3 Bagrut-focused creators for paid posts ($300–500 each).
- Ship Bagrut Intensive product page and checkout — live by June 1 for May/June exam season.
- Implement daily-streak + parent notification system for churn reduction.
- Deploy parent-targeted Facebook campaigns at $30–50 CAC to Israeli 40–55yo with high-school-age kids.

Days 61–90: Launch & Optimize

- Public launch: press release in Hebrew tech media (Calcalist, TheMarker). Reach 1,000 paid subscribers.

- A/B test paywall: with-CC vs without-CC trial. Target 18%+ trial-to-paid.
- Monitor and optimize: Bagrut Intensive attach (target 20%), trial-to-paid (target 22%), monthly churn (target 7%).
- Apply for Israeli Innovation Authority R&D grant ($100–150K, 12-month horizon).
- Begin Seed-round conversations with early-stage Israeli VCs (Pitango, TLV Partners, Grove Ventures).

Year-1 Milestones

Month	MAU	Paid	ARR	Cash Burn (monthly)	Key Gate
Month 3	500	100	$17K	$22K	Soft launch working, 5% trial→paid
Month 6	2,000	700	$96K	$35K	First Bagrut cohort complete, 18% trial→paid
Month 9	3,500	1,500	$207K	$42K	Family tier launched, ambassador program at 25 schools
Month 12	5,000	2,500	$343K	$45K	Seed round begins at 8.6K MAU / $1M ARR projection

Decision Triggers for Pivoting

Pivot to Architecture A (Freemium) if: trial-to-paid conversion stalls below 10% after 6 months of aggressive optimization, AND organic signup rate exceeds 500/week. (Signal: users love it but won't pay — monetize at scale with free acquisition.)
Pivot to Architecture C-adjacent B2B if: 3+ Israeli school districts actively request institutional licenses unprompted. Switch the business model, not the product.
Shut down if: by Month 9, MAU is below 1,500 AND monthly churn exceeds 12% AND Bagrut Intensive attach below 8%. These are three simultaneous failure signals — any two alone are recoverable.

Why the Founder Is Uniquely Suited

Ariel is 16, Israeli, a Bagrut student himself, and has logged 800–1,000 hours building X+1. He is the target customer, distribution channel (classmates + teachers who know him), and engineering team in one person. He speaks Hebrew natively, tests in real exam conditions, and has relationships with the school ambassadors the business needs. **No adult founder has this combination.** The founder's age is a strategic asset — he ships faster, costs less, and understands the user in ways consultants cannot replicate.

The Three-Year Story for Investors

Year 1 (2026): Launch Israel. 5K MAU, $600K ARR, 70% Israeli revenue. Validate Bagrut Intensive as profit center.

Year 2 (2027): Scale Israel + launch UK GCSE beta. 25K MAU, $3M ARR. Raise $3M Seed. Achieve break-even in Israeli unit economics.
Year 3 (2028): Expand to US AP + IB. 100K+ MAU, $10M+ ARR, break-even overall or modest profit. Raise $15–25M Series A.
Exit: Strategic acquisition by Khan Academy, Duolingo, Quizlet, Google (Photomath consolidation), or ByteDance (Gauth consolidation) at $75–200M — a 150–400× return on a $500K pre-seed.

Appendix — Methodology, Sources, Formulas

Core Formulas

Net Revenue Per Paid User (monthly)

For a blended tier mix, annual opt-in share, VAT geography, and payment fees:
Net ARPU = [Σ (tier_share × (monthly_share × price_monthly + annual_share × price_annual/12))]
× [PCT_IL × (1 / 1.18) + (1 − PCT_IL) × 1] // VAT adjustment
× (1 − 0.029) // Stripe percentage
− $0.30 × monthly_share − $0.30/12 × annual_share // Stripe fixed fee

Gross Profit and LTV

Gross profit per paid user per month = Net ARPU − Variable cost per paid user
LTV = Gross profit per paid user per month / Monthly paid-user churn rate

CAC Per Paid User

CAC (blended) = cost to acquire one new user via paid channels (weighted by channel mix)
Paid-channel share = 40% of new-user acquisitions (the rest organic/viral)
CAC per paid user = Blended CAC / Conversion rate (trial-to-paid for Sub, free-to-paid for Freemium)

Operating Profit

Operating profit = Net revenue − Variable cost (after credits) − Fixed cost − Team cost − Marketing spend

Key Assumptions (Base Case)

Parameter	Value	Source / Justification
Israeli VAT	18%	Israeli Tax Authority, consumer digital services 2026
Stripe fee	2.9% + $0.30	Stripe standard pricing, consumer
% Israeli users	70%	Beachhead strategy; shifts to 40% by Y3 with international expansion
% paid-channel acquisition	40%	Conservative; Duolingo reports 60%+ organic at scale
Paid-user monthly churn	8% (Sub), 9% (Free)	RevenueCat 2025; EdTech sector 9.6% avg
Free-user monthly churn	15%	Conservative for engaged free users with SRS retention
Trial-to-paid conversion	18%	Mid-range of RevenueCat 2025 hard-paywall 12–25% band
Free-to-paid conversion	4.5%	Between RevenueCat freemium 2.18% median and Duolingo 8.8% ceiling
Annual opt-in rate	22%	Mid-range per brief (15–25% consumer EdTech norm)

Bagrut Intensive attach	10–12%	Base case; optimistic 30–50% targeted
Startup credits (Y1 only)	$65K	Anthropic $25K + Vercel $30K + AWS $10K (verified April 2026)
Paid-user variable cost	$7.50–$8.00/mo	Aggressive Haiku routing (60% of actions Haiku-compatible)
Free-user variable cost	$1.00/mo	Tight caps + Haiku-only routing

Sources

Competitor Pricing & Metrics (verified April 2026)

Khanmigo pricing ($44/yr parents, $4/mo) — khanmigo.ai/pricing
Duolingo Q1 2025 Shareholder Letter (8.8% conversion, 9.5M paid subs)
Photomath ($9.99/mo, $69.99/yr) — Google-owned
Gauth (ByteDance) — 300M users, $18M/mo Q3 2025
Quizlet Plus ($7.99/mo, $35.99/yr, 75M MAU, ~$120M ARR)

Industry Benchmarks

RevenueCat 2025 State of Subscription Apps
Subscription Statistics 2025 (churn, conversion)
Business of Apps — Duolingo Revenue and Usage Statistics 2026
Chegg revenue decline and 45% layoffs (CNBC, Oct 2025)
TikTok CPM benchmarks 2025 (Varos / Trendtrack)

Startup Credits (verified April 2026)

Anthropic Startup Program — AICredits.co guide
Vercel for Startups ($30K credits, 2026 program)

Israeli Market

Haaretz — Israeli private tutoring industry
Naale Elite Academy — Israeli Bagrut & Yechidot system

Modeling Caveats

- All figures in USD unless noted. Israeli VAT embedded in list prices (pass-through to tax authority).
- Credits apply only to Year 1 variable cost — Year 2 cliff is explicit in each architecture.
- Fixed infrastructure cost scales sub-linearly (plan tier jumps). Actual cost at any given scale may vary ±20%.
- Marketing spend is modeled as 40% of churn-replacement new-user acquisition × blended CAC. Actual marketing may be higher during growth phases (front-loaded) and lower at scale (organic dominance).
- Founder compensation: $0 Year 1, $30K Year 2 (plus $12K part-time contractor). Excluded: equity dilution, legal, accounting.

- Sensitivity analysis holds all other variables constant; in reality variables correlate (price ↑ → churn ↑).
- No currency-hedging cost modeled; 70% of revenue denominated in NIS, 30% USD.
- Bagrut Intensive modeled at 10–12% attach (conservative); strategic target 30–40%.

What Is NOT Modeled

- Viral coefficient > 1.0. WhatsApp virality is treated as lower CAC, not as exponential growth.
- Seasonality explicitly. Bagrut Intensive amortized across 12 months; actual cash flow spikes in Q2.
- Feature-driven pricing-power changes. Interactive diagrams, TikZ walkthroughs, and parent portals may command premiums not captured in base case.
- Haiku 4.5 price reductions (shown in sensitivity but not base case).
- B2B school licensing revenue (shown in Architecture C but not added to A/B as incremental).